

Single-Cell RNA Sequencing Reveals Tumor Microenvironment Heterogeneity in Immunotherapy

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ABSTRACT

Tumor cellular heterogeneity and immune evasion mechanisms pose major obstacles to immune checkpoint blockade therapies. We perform high-throughput single-cell RNA sequencing (scRNA-seq) on patient tumor biopsies across multiple solid tumor cohorts. Transcriptomic profiling identifies distinct exhausted CD8+ T-cell subsets and immunosuppressive tumor-associated macrophages, highlighting predictive biomarkers for personalized oncology.

Keywords: Single-Cell RNA Sequencing, Cancer Immunotherapy, Tumor Microenvironment, T-Cell Exhaustion, Oncology

1. Introduction

While anti-PD-1 and anti-CTLA-4 immunotherapies have transformed oncology, response rates vary widely. Bulk tissue profiling masks critical cellular subpopulations driving therapeutic resistance within tumors.

2. Single-Cell Transcriptomic Profiling

We sequenced over 120,000 individual cells from metastatic melanoma and non-small cell lung cancer lesions. Unsupervised clustering revealed distinct cellular states across tumor, stromal, and infiltrating immune compartments.

3. Immunological Insights

A specific progenitor exhausted CD8+ T-cell population was strongly correlated with durable response to PD-1 blockade. Conversely, M2-polarized macrophages mediated local immunosuppressive signaling.

4. Conclusion

Single-cell transcriptomics provides high-resolution maps of tumor immune interactions, facilitating the discovery of synergistic combination immunotherapies and personalized predictive oncology biomarkers.